

ADS-508 – Team Project Status Update Form

Fill out this form and submit it by the end of Module 4 in Canvas.

Team Number: Group 6

Team Leader/Representative: Jun Sik Ryu

Full names of team members:

1. Jun Sik Ryu
2. Faye Shawntel Corprew

Title of your Distributed Data Science Pipeline project:

Predicting U.S. Housing Prices through Machine Learning Using Zillow Market Indicators

Short description of your distributed data science pipeline project and objectives:

Project goal is to develop an end to end distributed data science pipeline to predict median housing prices across U.S. metropolitan areas using Zillow market data. The pipeline integrates multiple datasets, including home value index, inventory, market temperature, and sale prices. Datasets are ingested into AWS S3 and processed using SageMaker.

Name of your selected dataset: Zillow Housing Market Data (ZHVI, Inventory, Market Temperature, Median Sale Price)

Description of your selected dataset (data source, number of variables, size of dataset, etc.):

The dataset is obtained from Zillow Research and includes multiple publicly available housing market indicators at the metropolitan level. It consists of several datasets, including the Zillow Home Value Index, housing inventory, market temperature index, and median sale prices. Each dataset contains geographic identifiers such as region and state, along with time series data recorded at monthly or weekly intervals. After transformation and merging, the final dataset contains approximately 230,000 observations with features representing supply, demand, and temporal dynamics. These variables enable the analysis and prediction of housing price trends across different regions and time periods.

Are you using and practicing GitHub as a code hosting platform for version control and collaboration? If yes, provide the link here: https://github.com/jsryu-git/ADS508_Final

How many times have your members met in the last two weeks? 3 times

List the specific contributions that each team member is providing for the Final Team Project in the table below.

- **NOTE:** ALL students on the team should contribute equally to the Final Team Project.

Team Member 1 (Jun Sik Ryu)	Team Member 2 (Faye Shawntel Corprew)
Project planning, dataset research, and dataset review	Implemented data ingestion and S3 integration
Data exploration and validation	Drafted assignment 3.1 rough draft_1
Developed data cleaning, transformation, and feature engineering code	Build machine learning model
Manage GitHub repository and version control	Assist with review and refinement of project deliverables
Documentation and interpretation of results	Presentation

Comments/Roadblocks: One of the primary challenges encountered was the high proportion of missing values in the median sale price variable after merging datasets with differing temporal granularities. To avoid introducing bias, missing target values were excluded rather than imputed, which reduced the usable sample size. Additionally, the project was constrained by a \$50 AWS Lab budget, requiring careful management of time and resources. Working with large datasets in S3 and SageMaker for the first time introduced time pressure and uncertainty regarding resource limits, which created some operational challenges but also encouraged more efficient workflow management and prioritization.